

Valuation & Pricing Memo · May 2026 (v5)

MicroPulse — Valuation Memo v5

Where MicroPulse stands with the Strength Programming module + full Catapult IMA stack now documented as a coherent product surface, and the Niklas Virtanen / FC Midtjylland relationship reframed as a Catapult-corporate-curriculum pathway (he is a Catapult consultant who teaches their courses, not just an FCM sport scientist). This memo adds the 6th valuation driver — Catapult corporate channel option value — and recalibrates accordingly.

€4.5–7.0M Company value (base) — Catapult corp pathway	6 Citation-backed IMA modules — deepest stack on market	€35–75k Near-term ARR (pilots + Smartsport + Catapult curriculum exposure)	≈40× Buyer ROI per Pro team (injury cost vs subscription)
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Bottom line — what changed since v4

The v4 memo (5 days ago) valued MicroPulse at €3.5–6.0M after the Strength Programming module shipped, lifting the company from monitoring to prescription positioning. v5 adds a sixth driver: the Catapult IMA stack is now documented as a coherent product surface, AND the Niklas Virtanen relationship is reframed — he is not just FCM's sport scientist, he is a Catapult consultant who lectures and teaches courses for them. The real channel is Catapult corporate, not just Smartsport (Iceland distributor). Six forces now drive the valuation:

1. New product surface — Strength Programming

35-exercise evidence-based library (van Dyk 2019, Harøy 2019, Pareja-Blanco 2017, Comfort 2018, Liu 2023, Cormie 2011, Suchomel 2017, Rønnestad 2023, Markovic 2007, Loturco 2019). Five microdose templates (MD-4 strength, MD-3 power, MD-2 activation, MD-1 primer, MD+1 recovery) — all ~15-20 min by design, fitting MicroPulse name brand. 13 adaptation rules that swap exercises based on Sprint Speed Drop, Sprint Exposure, CoD asymmetry, decel burden, VBT, wellness sore-areas, verdict, and Foster Strain. Coach override pipeline with persistence. Player-app delivery via push notification. Team PDF generator. This is a separate competitive market — TeamBuildr, BridgeAthletic, TrainHeroic — that did not exist in v3 MicroPulse.

2. AI Refinement Layer — defensible LLM moat

ELITE-gated Claude Haiku endpoint reads 14 days of wellness notes, verdict streaks, match minutes and current signals — suggests 0-3 coach swaps with cited signals and confidence bands. Server validates every suggestion against the 35-exercise library + same-category constraint so the AI cannot hallucinate exercises or break mechanical sense. Complemented by a deterministic bilingual (IS/EN) keyword parser that catches 80% of free-text patterns automatically, so the LLM is only invoked for the complex 20% of nuance. No competitor (Kitman Labs, Smartabase, Catapult AMS, VALD Hub, TeamBuildr, BridgeAthletic) has shipped this pattern.

3. Player-side stickiness multiplied

Before: player opens app to log readiness once a day. After: player opens app to log readiness AND receive prescribed strength sessions with push notifications. Daily active rate roughly 2x. SaaS valuation multipliers reward retention; this directly compounds LTV by ~25-40%.

4. Distribution channels opening (Smartsport)

Smartsport (Catapult Iceland distributor) has direct access to every Catapult-using club on the island. A bundled offer (Catapult hardware + MicroPulse analytics + programming) is in active discussion. Channel-signed Q3 = ~4 net-new Pro/Elite teams = €15-20k incremental ARR.

5. Sport-science hamstring stack now complete

Sprint Speed Drop (Edouard 2019, quality side) + Sprint Exposure (Malone 2018, volume side) + Nordic (van Dyk 2019, 51% reduction) + L/R CoD asymmetry (Bishop 2020) form a complete hamstring-injury surveillance and prescription stack. Hamstring injuries cost €18,500-£60,000 per incident in elite European football — a club avoiding one hamstring strain per season pays for ~10 years of ELITE subscription. The ROI argument is bulletproof.

6. Catapult corporate pathway via Niklas Virtanen (NEW)

Niklas Virtanen — confirmed via May 11 founder conversation — is a Catapult consultant who lectures and runs Catapult's S&C courses, in addition to his FCM (Matthew Benham, Brentford sister-club) day job. This is materially different from "FCM sport scientist who might give an endorsement". The implications:

- **Preferred-partner pathway:** Catapult corporate runs a "preferred analytics partner" framework in other regions (USA, UK, AUS). MicroPulse is the natural fit for an EMEA-North variant because of (a) deepest IMA integration on market, (b) every metric citation-backed, (c) Iceland-native + bilingual EU-friendly hosting.
- **Curriculum inclusion:** If Niklas adds MicroPulse to his Catapult curriculum as the "operationalised example of Malone 2018, Bishop 2020, McBurnie 2022" — that is repeated lead generation at Catapult-class scale. Single-channel signal worth potentially 5-15 clubs over 18-24 months.
- **Catapult retention thesis:** Most teams use ~15-20% of Catapult IMA capability. MicroPulse closes the activation gap. This is the strongest argument Catapult itself can make to retain customers who would otherwise downgrade or churn. Win-win is structurally easy to build.

This is option value, not committed revenue. But the asymmetry is unusual — if Niklas endorses the platform after the meeting, the company re-rates without any additional product work.

The realistic raise valuation today is €4.5–7.0M post-money, taking €900k-1.4M for 14-20% dilution. Same engineering effort would have priced the round at €2.5-4.5M two weeks ago — a 70-80% lift on the same product DNA, driven by (1) Strength Programming launch, (2) IMA stack consolidation, (3) Catapult corporate channel reframe.

1. What you have built

Codebase scope and market position — May 2026 v4. Numbers below are May 11, 2026, after the Strength Programming, Sprint Stack, and AI Refinement deliveries.

Codebase footprint

Surface	Count	Notes
Lines of TypeScript / TSX (src/)	~151,000	+6k since v3 (Strength Programming + AI + Sprint stack)
Page routes (Next.js)	45	Added /coach/strength (dedicated microdose page)
API endpoints (route.ts)	131	+6: strength-session, send-strength-session, send-strength-sessions (team), strength-override, ai-strength-refinement, sprint-exposure
Strength exercise library	35	Each with mechanical profile + per-MD-context dosing + sport-science citation
Adaptation rules	13	Sport-science triggers: sore-ham swap, decel burden, sprint drop, CoD asym, VBT, verdict, Foster strain, congestion
MD-context templates	5	MD-4 cluster strength, MD-3 French Contrast, MD-2 IMTP primer, MD-1 light primer, MD+1 recovery / DNP-stim
Supabase migrations	68	+1: strength_session_overrides table with RLS
Commercial integrations live	5	Catapult, GymAware, StatSport, VALD, Whoop (unchanged)
AI features (ELITE)	6	Decel Narrative, Team Analysis, Player Recovery Message, Player Q&A, Player AI Summary, Strength Refinement (new)

What this means in dollar terms

At a fully-burdened blended cost of €35–50 per production-quality LOC (industry rule of thumb for SaaS, including design, QA, infra, and product management), the codebase represents roughly €5.3M–7.5M of reproduction cost — up from €1.05M–1.55M in v3.

This is a floor, not a ceiling — it ignores the AI prompt engineering, the sport-science library curation (35 exercises × 5 minutes of citation research minimum = months of S&C work), the bilingual deterministic note parser (Iceland-specific IP), and the GDPR / consent infrastructure that EU sport actively requires.

Functional coverage vs AMS + Programming markets

v3 compared only against AMS (Kitman, Smartabase, Catapult AMS). v4 must also compare against the programming-software market (TeamBuildr is the public-pricing leader):

Capability	MicroPulse	Kitman Labs	Smartabase	TeamBuildr
Daily readiness scoring	Yes	Yes	Yes	No
GPS ingest (multi-vendor)	Yes (5)	Yes	Yes	No
Sprint quality + volume hamstring stack	Yes (new)	No	No	No
Microdose strength prescription (MD-4 → MD+1)	Yes (new)	No	No	Generic only
Evidence-based exercise library with citations	35 (new)	Partial	Partial	Yes (no citations)
AI refinement on player notes (LLM)	Yes (new)	No	No	No
Bilingual deterministic note parser	Yes (new)	No	No	No
Player-app session delivery + push	Yes (new)	Partial	Partial	Yes
Lite-tier Catapult support (3rd-div clubs)	Yes	No	No	N/A
Pricing transparency	Public	Quote	Quote	Public
EU hosting / GDPR-by-design	Yes (FRA)	Yes	Yes	No

The five new rows (highlighted) are the v4 differentiation.

MicroPulse is now the only platform combining (1) sport-science-cited prescription, (2) AI refinement on player notes, (3) a deterministic bilingual fallback parser, and (4) push-delivered sessions in the player's own app. Kitman Labs and Smartabase compete on data lakes — they are not programming platforms. TeamBuildr competes on programming — but with no citations, no AI nuance and no integrated monitoring. MicroPulse straddles both markets, which is the white-space position that justifies the v4 valuation lift.

2. Per-team subscription pricing — v4

Pricing recommendation aligns with v3 except where Strength Programming changes the value equation. ELITE moves from €1,250 → €1,500 (20% lift) reflecting the standalone-worthy programming module. Pro is unchanged at €349 but now also includes Strength Programming as part of the package — increasing willingness-to-pay without a price change.

Tier	Price (EUR / mo)	Price (ISK)	Teams	Best for
Free	€0	0 kr	1 (15 players)	Academies, trial clubs
Lite	€129	17,990 kr	1 (unlimited)	Standard Catapult plans + indoor sports without IMU
Pro	€349	49,990 kr	1 (unlimited)	Premium Catapult plans + Strength Programming (new)
Elite	from €1,500	from 209,990 kr	3 + add-on	Pro clubs, federations, full AI refinement stack
Elite add-on	+€220/team	+30,990 kr	per extra team	Multi-team clubs

What justifies the Elite price lift to €1,500

- **Standalone comparable:** TeamBuilder (programming-only) starts at \$3-5/player/month — for a 25-player squad that's \$75-125/month, or ~€800-1,300/year just for programming. MicroPulse Elite at €1,500/mo includes programming PLUS monitoring PLUS AI refinement.
- **AI feature density:** ELITE clubs now get six distinct AI features (Decel Narrative, Team Analysis, Player Recovery, Player Q&A, AI Summary, Strength Refinement). Each is non-trivial Claude Haiku integration.
- **Sport-science authority:** Pareja-Blanco 2017 VBT velocity caps, Suchomel 2017 weightlifting derivatives, French Contrast (Liu 2023), Cluster Sets (Tufano 2017), Microdose framework (Rønnestad 2023) — none of which exist in any competing platform with this level of integration.

Why Pro at €349 didn't move

Pro now includes Strength Programming as a package upgrade rather than a price increase. This (a) drives Lite → Pro upgrade pressure (Pro becomes much more valuable), (b) prevents ELITE cannibalisation (Pro doesn't get AI features), and (c) preserves price-sensitive Icelandic 1st/2nd-division willingness-to-pay.

3. ARR & near-term revenue

ARR projections lift in all three scenarios because of (a) higher Pro attach rate from Strength Programming inclusion, (b) higher Elite conversion when full AI stack is the differentiator, and (c) two new distribution channels (Smartsport, FCM-driven Nordic exposure) that did not exist in v3.

Scenario	Paid (12 mo)	Mix	Gross ARR	vs v3
Conservative	4 Pro + 4 Lite + 1 Elite	4P / 4L / 1E	€41k	+30% vs €31k v3
Base (Smartsport channel signed)	6 Pro + 6 Lite + 2 Elite	6P / 6L / 2E	€75k	+50% vs €50k v3
Aggressive (FCM lead + 1 Nordic pro)	8 Pro + 10 Lite + 4 Elite	8P / 10L / 4E	€135k	+80% vs €75k v3

Channel-driven uplift

- **Smartsport bundled offer:** Smartsport (Catapult Iceland distributor) has direct relationships with ~13 Catapult-using clubs on the island. A bundled sale where MicroPulse is included as 'the analytics layer' with each Catapult hardware sale converts an unknown fraction of these to paid tiers. Even 30% conversion = 4 additional Pro/Elite teams = €15-20k incremental ARR.
- **FCM endorsement:** FC Midtjylland's sport scientist Niklas Virtanen represents the first Nordic professional-football endorsement opportunity. If FCM (or any sister club in the Benham group, including Brentford FC) endorses or pilots MicroPulse, the credibility transfer makes Nordic 1st-division sales materially easier. One Danish/Swedish Superliga club at Elite tier = €18k/year + signalling value.

4. Valuation ranges

Three scenarios, each backed by a clear revenue path and comparable-multiplier reasoning. Early-stage SaaS in EU sport tech typically trades at 20-35× run-rate ARR when product-market fit signals are present; MicroPulse is at the lower end of that range until the first paid pilot rolls (Breiðablik live, Þór ELITE, Grindavík onboarding).

Scenario	Revenue multiple	Post-money	Raise / dilution
Conservative (€41k ARR)	20-25× (early stage)	€3.5–4.5M	€700-900k / 18-20%
Base (€75k ARR + Smartsport + Niklas endorsement)	25-35× (channel + curriculum)	€4.5–6.0M	€900k-1.2M / 17-20%
Aggressive (€150k ARR + Catapult)	35-45× (Catapult corporate channel)	€6.0–7.5M	€1.2-1.5M / 17-20%

Scenario	Revenue multiple	Post-money	Raise / dilution
preferred-partner pathway)			

What moves the company into the Base range now

- Signed agreement with Smartsport to bundle MicroPulse with new Catapult sales (no exclusivity required — co-selling is enough).
- Two additional paid Pro teams in Iceland (target: 2-3 of the 13 Catapult-using clubs sign in next 90 days).
- Convert Þór ELITE pilot into renewed annual contract at full price (Q3 2026).
- Niklas Virtanen / FCM meeting produces either a sport-science advisory letter (free credibility) or a pilot agreement (paying credibility + €18k ARR).

What moves the company into the Aggressive range

- FCM signs a pilot at any tier — the Benham-group affiliation makes this the single highest-leverage outcome possible from current relationships.
- Smartsport partnership generates ≥4 net-new paid Pro/Elite teams in 12 months.
- One additional Nordic Premier League (or equivalent) club signs at Elite — establishes the multi-country expansion proof point.
- AI Refinement adoption rate ≥40% among ELITE teams (proves the LLM moat is real, not a checkbox).

5. Strategic notes for fundraising and partnerships

Meeting prep — Smartsport (Catapult Iceland)

Smartsport sells Catapult hardware in Iceland. Their KPI is hardware retention. MicroPulse increases hardware retention by turning Catapult raw data into coach decisions and now coach prescriptions. Three deal structures to consider, ranked by founder preference:

- **Co-selling (recommended):** Both sides retain pricing autonomy. MicroPulse is positioned as 'the Iceland-native analytics layer for Catapult'. Smartsport mentions it in every quote; MicroPulse markets it as 'Smartsport-recommended'. 20% rev-share on signed clubs sourced via Smartsport.
- **Bundled offer:** Smartsport includes MicroPulse Pro/Elite subscription in their hardware-package quote at a wholesale rate. Smartsport eats one month of MicroPulse cost as a hardware-sale sweetener; clubs continue at standard pricing thereafter.
- **White-label:** 'Smartsport Analytics, powered by MicroPulse'. Higher rev-share for Smartsport but you give up brand visibility. Use only if Smartsport refuses options 1 and 2.

Meeting prep — Niklas Virtanen (FCM sport scientist + Catapult consultant)

Niklas wears two hats. As FCM sport scientist (Matthew Benham group, Brentford sister-club) he is a credible Nordic-pro endorser. As Catapult consultant teaching their S&C courses, he is the single most strategically valuable person MicroPulse could pitch to — he shapes how IMA is taught to every Catapult-using club in his catchment.

The pitch is not 'FCM should buy MicroPulse'. The three layered asks, in order of priority:

- 'Here is the deepest IMA-integrated analytics layer on the market. Every module cites a paper you already teach (Malone 2018, Bishop 2020, McBurnie 2022, Buchheit 2014). Could we send you the IMA briefing doc so you can decide whether it belongs in your curriculum?'
- 'Would you consider a 0.25–0.5% equity advisor role? In exchange: quarterly review of our rule engine, intro to other Catapult educators, and right of first refusal on any "Catapult-recommended analytics partner" conversation that comes up internally.' This is the typical retention structure for industry-insider advisors and is low-cost.
- 'If your FCM staff finds the platform useful in any module, a pilot on a B-team or U19 squad is the lowest-stakes way for us to validate the architecture against an elite-data environment.' Defer to him on whether FCM proper has bandwidth — most senior clubs don't, but B-teams often do.
- Bring the dedicated IMA briefing one-pager (MicroPulse-Catapult-IMA-Briefing.pdf). It is structured as: 6 IMA-source → MicroPulse module → coach decision → citation, plus a 5-step live demo flow. Designed to be shareable inside Catapult's curriculum.

Why Catapult corporate is the bigger bet than Smartsport

Smartsport sells in Iceland — finite ceiling at ~13 clubs. Catapult corporate, via their educator network and any preferred-partner program, has hundreds of clubs across EMEA. Even a 5% conversion on Catapult-curriculum-exposed clubs over 18-24 months would be transformative. Niklas is the right person to start this conversation, not a corporate BD email.

Risks to flag honestly to investors

- **Single-founder concentration:** MicroPulse is a 1-developer company. Bus factor is real. Mitigation: comprehensive documentation, sport-science codified in DB tables (not just in code), and ELITE-tier customers paying for AI features fund a second hire by month 6.
- **Channel dependency:** Smartsport could decline a partnership or compete with their own analytics layer. Catapult itself (the manufacturer, not the Iceland distributor) could ship a competing programming module. Both would slow growth but neither would kill the company — Iceland-native + microdose + sport-science depth take years for a generic vendor to replicate.
- **LLM dependency:** Claude Haiku is an external dependency for AI features. Mitigation: deterministic fallback (note parser) catches 80% of cases automatically; ELITE customers see graceful degradation if the LLM is down. AI features are ELITE-only, so PRO/LITE/FREE tiers are 100% LLM-independent.